

PC5215 Numerical Recipes with Applications

Midterm test, closed book, 1 hour 45 minutes, Thursday 29 Sep 2022

1. Given each of the following Python inputs to the interpreter, what will be the response:

- `1 + 0.5`
- `7 // 2`
- `(-7) % 3`
- `2 ** 4`
- `str="AB"; cat="CD"; str+cat`
- `{1,2,3} & {2,5,7}`
- `lis = [1,2,3]; lis[-1]`
- ```
def incr(x):
 x += 1
x=0; incr(x); x
```
- ```
def incrlis(x):  
    x[0] +=1  
x=[0]; incrlis(x); x
```
- `zeros = [0,0]; A = [zeros, zeros]; A[0][1] = 1; A`

2. The IEEE representation of single precision floating point numbers consists of 32 bits, the highest bit s is for the sign, and the next 8 bits are for the biased exponent e with a bias of 127, and the rest 23 bits are for the fractional part f (mantissa). Ignoring the possibility of denormalized numbers, the bit pattern represents the number $(-1)^s(1 + f) \times 2^{e-127}$.

- Work out the bit pattern in hexadecimal notation (0 to 9, A to F) for the value 5.5.
- What is the smallest and largest positive single precision floats based on the formula give above?
- In physics, many constants are very small. For example, the Planck constant h in SI units is $6.626 \times 10^{-34} \text{ m}^2 \text{ kg/s}$, and square of h will be even smaller (such as in the calculation of the kinetic energy of a quantum particle). Explain or argue based on your answer of part b if it is a good idea to use SI units in numerical calculation involving the Planck constant. State the merit and drawback of using SI units. Will your conclusion change if you use double precision instead (which is the default in Python)?

3. In physics we often encounter tri-diagonal matrices such as a chain of connected springs in one dimension. A tri-diagonal matrix A_{ij} has zero elements if the row and column index

differs by more than 1, something like
$$\begin{pmatrix} b_1 & c_1 & 0 & \dots & 0 \\ a_2 & b_2 & c_2 & 0 & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & 0 & a_{N-1} & b_{N-1} & c_{N-1} \\ 0 & \dots & 0 & a_N & b_N \end{pmatrix}$$
. The LU

decomposition of such a tri-diagonal matrix can be done more efficiently than the standard Crout's algorithm implies, which is $O(N^3)$ for an $N \times N$ matrix.

- Give the original α, β equations, where α is the lower triangular matrix with diagonals α_{ii} equal to 1, and β is the upper triangular matrix, such that $\alpha\beta = A$, ignoring pivoting.

- b. Now taking into account that A is tri-diagonal, show that the LU factorization is also tri-diagonal, i.e., α matrix only have the diagonals of 1, and one lower off-diagonal entries not zero, and β is diagonal plus one upper off-diagonal nonzero.
 - c. Based on the result of part b, give the computational complexity, i.e., the power k in $O(N^k)$ when performing the special tri-diagonal LU decomposition.
 - d. Implement the method given in part b as a working Python code (without pivoting).
- 4. Consider a three-point polynomial interpolation problem with Neville's algorithm for the P-tableau. The input data $(x_i, f_i), i = 1, 2, 3$ are $(0, f_1), (1, f_2), (2, f_3)$. Note that the abscissa x_i are equally spaced.
 - a. The independent variable x is given as a symbol, rather than a specific value. Go over the Neville's algorithm to determine $\lambda_{12}, \lambda_{23}, \lambda_{13}$, and then compute the polynomials $P_{12}(x), P_{23}(x), P_{123}(x)$.
 - b. Using the quadratic polynomial $P_{123}(x)$ determined in part a, show that the integral from 0 to 2, $\int_0^2 P_{123}(x)dx = w_1 f_1 + w_2 f_2 + w_3 f_3$, is given by the Simpson rule. Determine the weights $w_i, i = 1, 2, 3$.
 - c. Check and show that the above integration formula (when P_{123} is replaced by some arbitrary function f of x) is also exact for x^3 , but not for x^4 .

- The end -